CDO Valuation with the Lintner Model

Mini-Project 3, Part 2 | FRE 6103 Valuation for Financial Engineering, NYU Tandon

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Interactive page:

`https://pawarraj8888.github.io/valuation-assignments/cdo-analysis/valuation.html` (change Class B's principal and coupon and the values update)

Aim: To value Class A, Class B and the bank's residual with the Lintner model from class, using the cash flows of Part 1 in 1000 cases, to find the cost of risk implied by the bonds' market price, and (bonus) to find the structure that gives the residual the most value with Class A held at $15 MM.

1. Key Assumptions

Collateral and Defaults	As in Part 1: 10 bonds of $10 MM, 6% coupon paid quarterly, 5 years; BIS debt model with a 4% annual default probability, 60% LGD and 0.20 correlation between default dates; the same 1000 moment-matched cases (seed 6103).
Classes	Class A: $15 MM at 4.5%. Class B: $30 MM at 5.5%. We use the target rates in the assignment as the coupons, so a class sold at par pays its target rate. Both are bullet bonds: Class A is owed $0.16875 MM a quarter and $15.16875 MM at maturity, Class B $0.4125 MM and $30.4125 MM. If no bond defaults the residual receives $0.91875 MM a quarter and $55.91875 MM at maturity. The waterfall is the same as in Part 1: each quarter stands alone, so a shortfall is not made up later.
Risk-Free Rate	4% a year, compounded quarterly: a payment in quarter k is discounted by (1 + 0.04/4)^-k.
Bonds' Market Value	The pool's promised cash flows discounted at the 9% market yield, compounded quarterly: $88.03 MM for the ten bonds.
Lintner Method	From Lecture 6. Cost of risk = expected PV of the pool at the risk-free rate less the bonds' market value. Cost per unit risk = cost of risk / variance of the pool's PV. Value of a tranche = its expected PV less the cost per unit risk times the covariance of its PV with the pool's PV. Covariances use the sample (n - 1) formula, like COVARIANCE.S in Excel; the choice cancels out of the values.
Default Indicator	For each case and class, 1 if the class is paid less than it is due in some quarter, otherwise 0.
Units	$ millions. All PVs are as of today, at the risk-free rate.

2. Implementation Steps

1. **Scheduled Cash Flows:** Promised quarterly payments to Class A and Class B from their principal and coupon, then the Part 1 waterfall on the pool's cash flows in every case: Class A first, then Class B, the rest to the residual.
2. **1000 Simulations:** The 1000 cases come from the fixed random numbers of Part 1 after moment matching, so the default dates, and therefore the pool's cash flows, are exactly the ones in Part 1.
3. **PVs, Covariances and Default Indicators:** In every case we discount each tranche's 20 quarterly cash flows at 4%. That gives a 1000 x 3 table of PVs (Class A, Class B, residual), its 3 x 3 covariance matrix and a default indicator per case for each class.
4. **Cost of Risk:** The pool's expected PV at 4% is $98.50 MM, but the bonds trade for $88.03 MM, so the market charges $10.47 MM for their risk. The pool's PV has a variance of 99.43 across the cases, so the cost per unit risk is 10.4735 / 99.43 = 0.10533.
5. **Lintner Values:** Each tranche is charged the cost per unit risk times its covariance with the pool. The three covariances add up to the pool's variance, so the three values add up to the bonds' market value.
6. **Cross-check:** The Valuation sheet of the Excel workbook does the same steps with formulas. It gives the same PVs in all 1000 cases and the same values as the Python code.

3. Values

	Principal	Expected PV at 4%	Max error (2 s.e.)	Covariance with pool	Risk charge	Lintner value	Value per $100	Cases short
Class A	15.00	15.34	0.00	0.00	0.00	15.34	102.26	0
Class B	30.00	32.03	0.00	0.00	0.00	32.03	106.77	0
Residual	-	51.13	0.63	99.43	10.47	40.66	-	-
Pool (bonds)	**100.00**	**98.50**		**99.43**	**10.47**	**88.03**	**88.03**	

Table 1. Lintner values of the Part 2 structure, $ MM, from 1000 cases. The max error is two standard errors of the expected PV, as in class, from the formula for independent cases; with moment matching that formula overstates the error (see Precision). Covariance with the pool is the sum of the row of Table 2. Risk charge = 0.10533 x covariance. For the pool row, the value per $100 column shows the bonds' market value.

Neither class misses a payment in any of the 1000 cases: the default indicators are 0 everywhere. Each quarter the pool pays at least $0.60 MM, more than the $0.58 MM the classes need. At maturity they need $45.58 MM, and the pool pays less than that only if all ten bonds default (it would then pay $40.60 MM). That has a 0.02% chance under the model and happens in none of our cases: the worst case has 9 defaults and still pays $46.69 MM at maturity. So the PVs of Class A and Class B are the same in every case, their covariances with the pool are 0, and they carry no risk charge: each is worth its promised cash flows at the risk-free rate. All of the $10.47 MM cost of risk falls on the residual, which is worth $40.66 MM against an expected PV of $51.13 MM.

Because the classes are never short in our cases but pay more than 4%, they are worth more than their principal: Class A $15.34 MM for $15 MM, Class B $32.03 MM for $30 MM. If the bank sells them at par, the investors receive $2.37 MM more than they pay. The bank keeps a residual worth $40.66 MM after putting in $43.03 MM of its own (the bonds less the $45 MM raised), a loss of $2.37 MM. The fair coupon for both classes is 4.0%, the risk-free rate; at that coupon each class is worth exactly par and the residual is worth $43.03 MM.

Covariance of PVs	Class A	Class B	Residual	Sum (with pool)
Class A	0.0000	0.0000	0.0000	0.0000
Class B	0.0000	0.0000	0.0000	0.0000
Residual	0.0000	0.0000	99.4309	99.4309

Table 2. Covariance matrix of the three PVs across the cases. Class A and Class B have the same PV in every case, so their rows and columns are 0.

We kept the assignment's structure, and with it neither class carries any risk, so the covariance matrix has only one entry that is not 0. To see the method charge a class for risk, Table 3 values the same cases with Class B at $60 MM (still at 5.5%). Class B is then short in 82 of the 1000 cases, its PV moves with the pool's, and it is charged $1.80 MM for risk: it is worth $61.62 MM against an expected PV of $63.43 MM. The residual's charge falls to $8.67 MM, and the three charges still add up to the same $10.47 MM.

	Principal	Expected PV at 4%	Covariance with pool	Risk charge	Lintner value	Value per $100	Cases short
Class A	15.00	15.34	0.00	0.00	15.34	102.26	0
Class B	60.00	63.43	17.14	1.80	61.62	102.71	82
Residual	-	19.73	82.30	8.67	11.06	-	-

Table 3. The same valuation with Class B at $60 MM and 5.5%, Class A unchanged, $ MM.

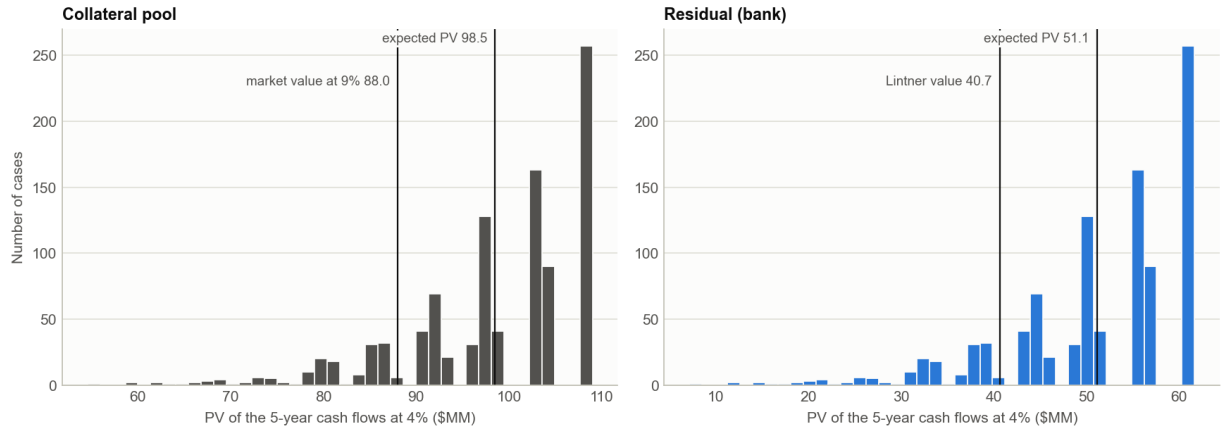


Figure 1. PV at the risk-free rate across the 1000 cases. Left: the pool, with its expected PV and the bonds' market value; the gap is the cost of risk. Right: the residual, with its expected PV and its Lintner value; here the gap is the whole risk charge.

4. What Moves the Values

Input	Value	Expected pool PV	Cost of risk	Cost per unit risk	Class A	Class B	Residual	B short
Correlation	0.0	98.45	10.43	0.1984	15.34	32.03	40.66	0.0%
Correlation	0.2 (base)	98.50	10.47	0.1053	15.34	32.03	40.66	0.0%
Correlation	0.4	98.40	10.37	0.0638	15.34	31.98	40.71	0.3%
Correlation	0.6	98.54	10.51	0.0472	15.34	31.85	40.84	1.3%
Correlation	0.8	98.72	10.70	0.0343	15.34	31.48	41.21	4.9%
Default prob.	2%	103.48	15.46	0.3076	15.34	32.03	40.66	0.0%
Default prob.	4% (base)	98.50	10.47	0.1053	15.34	32.03	40.66	0.0%
Default prob.	6%	93.72	5.70	0.0408	15.34	32.03	40.66	0.0%
Default prob.	8%	89.18	1.15	0.0069	15.34	32.02	40.67	0.2%
Default prob.	12%	81.46	-6.57	-0.0347	15.34	32.04	40.65	1.4%
LGD	40%	102.01	13.98	0.3164	15.34	32.03	40.66	0.0%
LGD	60% (base)	98.50	10.47	0.1053	15.34	32.03	40.66	0.0%
LGD	80%	94.99	6.97	0.0394	15.34	31.85	40.84	1.7%
LGD	100%	91.49	3.46	0.0125	15.33	31.45	41.24	3.3%

Table 4. Values with one input of the default model changed, the bonds' market value held at its 9% price. B short = share of cases in which Class B is paid less than it is due.

Changing an input of the default model moves the pool's expected PV, but the bonds' market value stays at $88.03 MM because it comes from their price, not from our model. So the cost of risk takes up the change and the values hardly move: the residual stays near $40.7 MM in every row, and the classes keep their risk-free values until they start to miss payments (Class B at a high correlation, default probability or LGD; Class A only at 100% LGD, in 0.1% of the cases). Correlation leaves the expected PV alone but spreads the pool's PV out, so the cost per unit risk falls from 0.198 to 0.034. At a 12% default probability the pool's expected PV ($81.46 MM) is below the market value, the cost of risk turns negative (minus $6.57 MM) and the method stops making sense: a 9% yield would be too low for bonds that risky.

Precision. We valued the structure again on 500 fresh sets of 1000 cases, once with the numbers as drawn and once after moment matching. Moment matching cut the spread of the pool's expected PV from $0.33 MM to $0.12 MM and the spread of the cost per unit risk from 0.0089 to 0.0041. The Lintner values moved even less (about $0.01 MM for the residual either way), because the bonds' market value pins the total and the classes are nearly always paid in full.

5. Bonus: Choosing Class B

We held Class A at $15 MM and 4.5% and searched over Class B's principal ($0 to $100 MM in $1 MM steps, 101 sizes) and coupon (0% to 10% in 0.25% steps), re-running the waterfall and the Lintner valuation on the same 1000 cases for each structure. The cost per unit risk does not change from one structure to another, because it comes from the bonds alone.

Taken literally, the residual is worth the most with no Class B at all: $72.69 MM. That follows from the method. The values of Class A, Class B and the residual always add up to the bonds' $88.03 MM, so whatever Class B is worth comes out of the residual, and the residual is worth the most when Class B is worth nothing. We held Class A's 4.5% coupon in the search because the assignment fixes Class A; a lower Class A coupon would move value to the residual in the same way. It is not a useful answer, because it ignores the cash the bank receives for Class B.

The bank's gain. So we measured each structure by the bank's gain: the cash from selling both classes at par plus the value of the residual it keeps, less the bonds it puts in (Figure 2 and Table 5). At Class B's 5.5% target rate the gain is negative for every Class B up to about $66.5 MM. Up to $65.5 MM, Class B's fair rate is under 5.5%, so the coupon pays its investors more than their risk is worth; at the proposed $30 MM the bank gives away $2.37 MM. At $65.5 MM, 5.5% is exactly a fair rate (Class B is short in 15.9% of the cases), and the bank still loses what Class A's 4.5% costs. Above about $66.5 MM the gain turns positive, but only because Class B investors would then pay more than the class is worth, which a buyer using the same model would refuse.

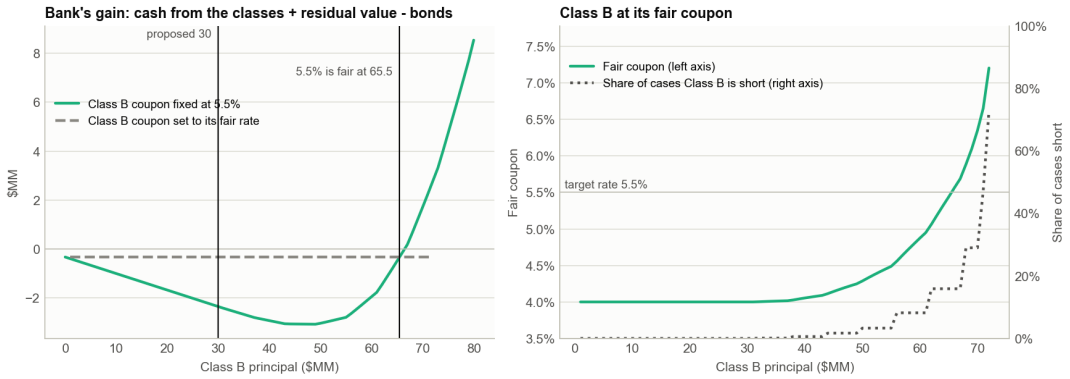


Figure 2. Left: the bank's gain against Class B's principal, with Class B's coupon fixed at its target rate and with it set to its fair rate. Right: Class B's fair coupon and the share of cases in which it is short at that coupon.

Class B principal	Class B value at 5.5%	Residual at 5.5%	Bank's gain at 5.5%	Cases short at 5.5%	Fair coupon	Residual at fair coupon
$10 MM	10.68	62.01	-1.02	0.0%	4.00%	62.69
$20 MM	21.35	51.34	-1.69	0.0%	4.00%	52.69
$30 MM	32.03	40.66	-2.37	0.0%	4.00%	42.69
$40 MM	42.61	30.08	-2.94	0.6%	4.05%	32.69
$50 MM	52.71	19.98	-3.05	3.3%	4.29%	22.69
$60 MM	61.62	11.06	-1.96	8.2%	4.87%	12.69
$65 MM	65.16	7.53	-0.50	15.9%	5.44%	7.69
$70 MM	67.96	4.73	1.70	29.0%	6.35%	2.69
$80 MM	71.11	1.57	8.55	74.3%	none	-

Table 5. Class B sizes with Class A held at $15 MM and 4.5%. Bank's gain = cash from selling both classes at par + residual value - bonds' market value. 'none': no coupon makes a Class B that large worth par.

Fair coupons. If Class B's coupon is set to its fair rate instead (the coupon at which it is worth exactly par), the bank loses the same $0.34 MM at every size, which is what Class A's 4.5% coupon costs. That holds up to about $72.8 MM, the largest Class B that any coupon can make worth par. That is a little more than the bonds less Class A ($72.69 MM), because the Lintner model gives the cases with no default a negative weight: at very high coupons the residual is worth slightly less than zero (as low as minus $0.14 MM in our search), the negative values the lecture warns about. The fair rate stays at 4.0% while Class B is never short (up to about $31 MM) and rises to 4.3% at $50 MM and 6.3% at $70 MM.

What this means. Once the classes are priced fairly, the size of Class B does not change the value of what the bank keeps compared with what it puts in: a larger Class B means less of the bank's own money in a riskier residual, but no gain. The residual can only gain at the classes' expense, by paying them less than a fair coupon. With the default indicator we can see how far that would have to go: if Class B investors took 5.5% as long as the class is short in no more than a set share of cases, the bank's best Class B would be $0 (no Class B) for any limit up to 10%, and $67 MM (gain $0.17 MM) with a 20% limit.

Our structure. Keep Class A at $15 MM and Class B at $30 MM, but pay both 4.0%, their fair rate, since neither is ever short. The residual is then worth $43.03 MM instead of $40.66 MM, the same as the $43.03 MM the bank puts in. The bank can take a larger Class B at its fair coupon to tie up less of its own money; under the Lintner model that does not add value.